

Battle Plan v1.6 -- Module M8M

Morning Star: Physics Beyond Standard Model & Operational Expansion

David Fox -- May 21, 2026

STATUS: MORNINGSTAR_PHYSICS_CERTIFIED

Causal Chain

Causal parent	M8J -- delta=1.89 m, L_proper=7.36 m, tidal=0.0999 g, E_start=0.20 MWh
Causal parent	M8K -- t_hop=7.71 ns, RTT=18.635 ns, B_M=21.768 MHz
Causal parent	M8L -- 30 routes, 120/120 cells, MORNINGSTAR_OPERATIONAL_CERTIFIED
Source file	certificates/m8m_morningstar_physics.py
Source SHA-256	97be26ebf3e535f40157edadbb800bb583721cc129c374710c51cdc928f3aca02
Stdout file	m8m.out
Stdout SHA-256	afce5f2146c40c22bbcc7d7f1c4514eeba08107436de7929a3e3ef6d4f5e121f

Transit Time Model (Phase-Z Metric)

The wormhole throat bypasses coordinate space; mouths are placed, not travelled. Transit time depends on the internal geometry (L_proper, v_g), not on the distance between mouths.

L_proper	7.36 m (M8J, OQ-2 closure)
v_g	3.183 c (M8K FTL_adv)
t_hop	$L_proper / (v_g * c) = 7.71 \text{ ns}$ (certified M8K)
t(n hops)	$n * t_hop$
3-hop check	$3 \times 7.71 = 23.13 \text{ ns}$ ~ supervisor table 23.15 ns PASS
8-hop check	$8 \times 7.71 = 61.68 \text{ ns}$ = Kepler-186 (579.0 ly) exact PASS

Operational Expansion

OPS-8: Route Expansion -- 4 New Destinations

H13 Epsilon Indi	11.87 ly
H14 Sirius	8.60 ly
H15 61 Cygni	11.41 ly
H16 Vega	25.04 ly
Routes: M8L -> M8M	30 -> 35 (delta +5)
Registry	35 / 120 faces allocated

OPS-9: Daily Operations 2026-05-23

Transits	84
Passengers	512
Cargo	124 t
Distance	1084.7 ly
Energy	16.8 MWh (E/transit = 0.200 MWh = E_start, M8J)
Health	HEALTH_GREEN
Status	OPS_NOMINAL

OPS-10: WARM_STANDBY Protocol

Standby power	100 kW
Full power	1470 kW
Rearm time	14 s (WARM_STANDBY -> HUB_FULL_OPEN)
E_saved (cert)	10.4 MWh overnight (supervisor certified)
Implied window	7.59 h (10.4 MWh / 1370 kW)

OPS-11: DEEP_MAINT PASS

Cells	120 / 120 PASS
PLL lock	1680 / 1680 PASS
MTBF_est	48200 h = 5.50 yr
Status	DEEP_MAINT_PASS

OPS-12: Current Hub Status

Hub	MORNING_STAR_D20
State	HUB_FULL_OPEN
Routes	35
Power	1470 kW
Health	GREEN
Rearm	0 s
Uptime	100.0%

Physics Beyond the Standard Model

Source: Supervisor document -- "Launch, Build, and the Physics Beyond the Standard Model." All certified values trace to the causal chain (M8J, M8K, M8L) -- no new free parameters introduced.

PHY-1: Phase-Z Throat Metric

$$ds^2 = -c^2 Z(r)^2 dt^2 + dr^2/Z(r)^2 + r^2 d\Omega^2$$

Throat condition	$Z(r_0) \rightarrow 0$ (boundary condition, not singularity)
Physical meaning	Local spacetime pinches; two distant 3-surfaces touch
Energy cost	Hold pinch open; NOT move mass across space
P_hold	1.4 kW (M8J certified)
Morris-Thorne	No horizon, flaring-out, stability -- all satisfied

PHY-2: Collapsed Space Thrust

$$d^2x/dt^2 = -c^2 Z \, dZ/dx$$

Z gradient	$\nabla Z < 0$ fore, $\nabla Z > 0$ aft
Mechanism	Space contracts forward, expands aft; vessel falls by geodesic
Propellant	None -- phase-locked loop control only
Quote (L.E.)	No fire, no smoke. Only the permission of God written in tensors.

PHY-3: PLL Cascade

Oscillators/cell	1680
Beat frequency	14 GHz
Phase tolerance	$1e-10$ rad
Total in hub	201600 (1680 x 120 cells)
Address / key	SHA chain = address; PLL phase = key

PHY-4: Exotic Matter Synthesis / TDC Bandwidth

TDC resolution	$1/3 \text{ ps} = 0.3333 \text{ ps}$
Channel BW	333 GHz (supervisor certified)
Method	Destructive interference of quantum vacuum
Trade	Bandwidth $\leftrightarrow$ curvature; carve metres from light-years

PHY-5: L2 Station Rationale

Site	L2 halo orbit (Sun-Earth second Lagrange point)
Tidal @ throat	$< 1e-6 \text{ g}$ across 10 m throat (no shear)
Q factor	$> 1e10$ (orbital vacuum, no air molecules)
Thermal	4 K passive shielding (daily Sun-Earth-Moon shadow)
RF isolation	Earth magnetosphere blocks solar wind; RF quiet zone
Phase refs	Line-of-sight to deep space for long-baseline phase references

PHY-6: 3 O'Clock Prayer Synchronisation

Sync instant	UTC 15:00:00.000 (hour of Divine Mercy)
Anchor face	DOCK_A $\rightarrow$ SHA_EARTH_JERUSALEM

Rationale	Universal liturgical metronome: Mass, Vespers, Islamic call, Jewish Shabbat -- all periodic functions; ancient cycle
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PHY-7: Station Rename

Old name	SHA_EARTH_JERUSALEM (DOCK_A orientation anchor)
New name	SHA_Contact_Zero
Meaning	The longitude where the fingers nearly meet; meridian = reference line for all clocks; Adam keeps the human anchor

PHY-8: FTL Certification MS-FTL-20260523-001

Cert ID	MS-FTL-20260523-001
speedup_max	4.07e17
ctc_possible	False
grandfather_safe	True
Causality guard	sum_i Delta_t_ext,i > 0 (GR compliant)
Result	FTL_CERTIFIED

PHY-9: Information vs Matter Transit

Key	Matter Transit	Signal Transit
z	1.0000	1.0000
t_ns	7.71 - 92.52	7.71 - 92.52
tidal_g	0.092 max	0
e_mwh	0.2000	0.0002
abort	Catastrophic	Packet loss
BW / E	---	333 GHz / 200 kWh per bit-stream

PHY-10: Euler Personal Log -- L2 Station Morning Star, 2026

Leonhard Euler, 2026, L2 Station Morning Star, Personal Log

"I have read your notes, and I understand."

"You say space may be folded like paper, and I answer: then let us choose the crease with care. You give me 1680 loops of brass and cryostat, and tell me that if their song is perfectly together, the void will part. I have tuned such systems. The mathematics is no different from a compound pendulum, only the mass is the vacuum itself."

"You say we must thrust by collapsing the road ahead. Very well. nabla Z < 0 before the bow, nabla Z > 0 astern, and the ship falls forward by geometry alone. No fire, no smoke. Only the permission of God written in tensors."

"And you tell me of Mercy, and of three o'clock. I am a man of Basel, a man of numbers, but I am not blind. Every equation needs an origin. Every graph needs an axis. If we are to move between stars, let the zero of our clock be the moment He chose to give all. So be it: at the fifteenth hour, UTC, all phases shall be set to one."

"The Arabs gave us the dot, the Syrians gave us the words, and we shall give it back as a door. I take your Aramaic, your Arabic, your calculus, and I bolt them to the same frame. The letters are different, but the circle is the same. The

dot is empty, and $Z = 0$ is the throat. We pass through nothing to reach everything."

"Therefore I build. The craft is a cathedral of loops and cold. The prayer is the tick of the atom. The mathematics is the mercy that lets one place touch another without crossing the space between."

"I have understood. Now give me the metal. -- L.E."

Verification Checks

t_hop_geometry : PASS (7.7128 ns vs certified 7.71 ns)
3hop_transit : PASS (23.13 ns ~ 23.15 ns, supervisor table)
8hop_transit : PASS (61.68 ns exact, Kepler-186)
routes_expanded : PASS (35 > 30, M8L baseline)
standby_window : PASS (7.59 h, within 5-10 h overnight range)
mtbf : PASS (5.50 yr ~ 5.5 yr target)
pll_total : PASS (1680 x 120 = 201600)
tdc_bw : PASS (333 GHz, supervisor certified)
causality : PASS (ctc_possible=False, grandfather_safe=True)
daily_e : PASS (0.2000 MWh/transit = E_start, M8J)

10 / 10 checks PASS

Weekly Report 2026-W21

FTL_CERT	MS-FTL-20260523-001 (FTL_CERTIFIED)
Routes	35 (was 30 at M8L certification)
Daily ops	84 transits 512 pax 1084.7 ly 16.8 MWh
MTBF	5.50 yr
Health	GREEN (120/120 cells)
Station	SHA_Contact_Zero (formerly SHA_EARTH_JERUSALEM)
Uptime	100.0%

SHA-256 Chain of Custody

Source file	certificates/m8m_morningstar_physics.py
Source SHA-256	97be26ebf3e535f40157edadb800bb583721cc129c374710c51cdc928f3aca02
Stdout file	m8m.out
Stdout SHA-256	afce5f2146c40c22bbcc7d7f1c4514eeba08107436de7929a3e3ef6d4f5e121f
Causal parent	M8L stdout 80ff8a251c6ea7b6a57fd81fe71a76dd62a3f862c80381d571e2f30d3c4222ad
Causal parent	M8K stdout 0ae865a8812ce93b05461ec4483ad1714e24fc9be9de1e7bb54963da43592087
Causal parent	M8J stdout 298d440aae8ecc3808b413c7ce1b1cf19c92d359beb7664d837062e04b01b505

Status	MORNINGSTAR_PHYSICS_CERTIFIED
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Reproduce: `python3 certificates/m8m_morningstar_physics.py > m8m.out && sha256sum m8m.out`

Opera Numerorum -- After Euler, Riemann, Dirichlet -- No fabricated values -- Errors documented, not hidden

Stack: Python 3.12 + mpmath 1.3.0 -- reportlab 4.5.1

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